UNDERSTANDING STARS

MASSIVE BALLS OF HOT, IONIZED GAS, STARS ARE POWERED BY NUCLEAR REACTIONS

Our galaxy contains hundreds of billions of stars—and there are hundreds of billions of galaxies, each containing similar numbers of these huge balls of plasma (hot, ionized gas). A star glows because it is hot, and most of the heat is generated by nuclear reactions in the star's core.

Around 6,000 stars are visible to the naked eye in the night sky. Apart from the Sun, they are so far away that, despite their enormous size, they appear only as tiny points of light, even through powerful telescopes.

THE SUN IS A STAR

The Sun is by far the closest star: the light and other radiation it produces takes eight minutes to reach Earth, compared with over four years from the next nearest star. Like other stars, the Sun is composed mostly of hydrogen and helium, with small amounts of other elements. Its luminous surface (photosphere) is white hot, with a temperature of about 10,000°F (5,500°C), and its outer atmosphere, the corona, is much hotter. The Sun is about 5 billion years old, and is about halfway through its life cycle.

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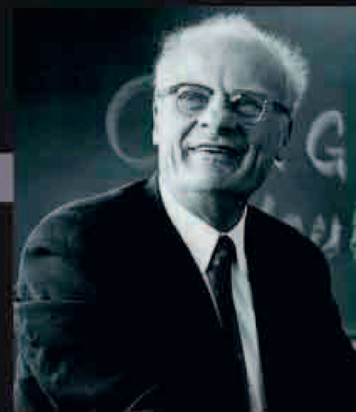
THE NUMBER OF **TIMES GREATER** THE SUN'S DIAMETER IS **COMPARED** WITH THE **EARTH'S**

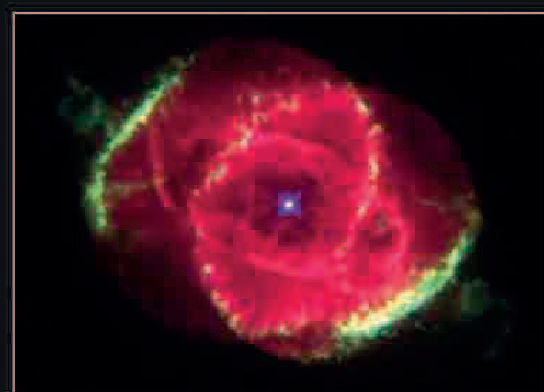
LIFE CYCLES OF STARS

Stars form in huge masses of gas and dust called molecular clouds. Gravity causes matter in denser regions of these clouds to clump together to form protostars. This gravitational collapse produces heat, which causes atoms to lose electrons, becoming ions, so the matter in the protostar becomes plasma—a mixture of ions and electrons. At the protostar's center, the high temperature and pressure cause nuclei of hydrogen atoms to fuse together to form nuclei of helium and some heavier elements. This nuclear fusion reaction releases energy, which heats the protostar further: a star is born. When the hydrogen runs out, nuclear fusion ends, and the star cools and collapses under its own gravity. A star's final destiny depends upon its mass; the most massive stars end up as black holes (see opposite).



HANS BETHE
In the 1930s, German-born physicist Hans Bethe (1906–2005) worked out how nuclear fusion builds elements inside stars, for which he was awarded the 1967 Nobel Prize in Physics.

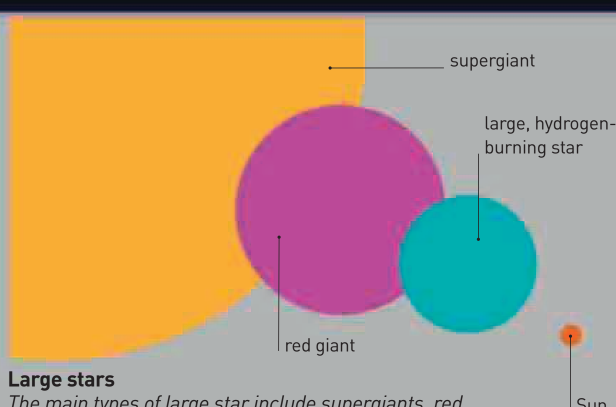
STAR BIRTH
The molecular cloud in the Carina Nebula (part of which is shown in this image from the Hubble Space Telescope) is one of the largest known regions of star birth in our galaxy, the Milky Way.



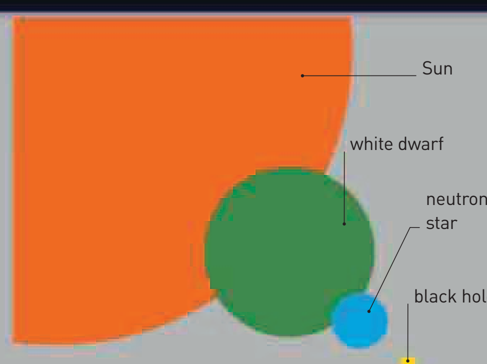
STAR DEATH
As stars of low to intermediate mass near the ends of their lives, they eject haloes of hot gas, forming objects known as planetary nebulae. At the center of each such nebula is a small remnant of the once much larger star, called a white dwarf.

STAR SIZES

Stars come in a huge variety of sizes. Supergiants, among the largest stars, can be over 1,500 times bigger than the Sun. The Sun itself has a diameter of about 870,000 miles (about 1.4 million km)—roughly average for a star in the main part of its life. The smallest stars, neutron stars, are only about 12.5 miles (20 km) across.



Large stars
The main types of large star include supergiants, red giants, and large hydrogen-burning stars. The Sun is an average-sized hydrogen-burning star.



Small stars
Small stars result from the death of larger stars. Stars like the Sun become white dwarfs, while more massive stars become tiny neutron stars or even black holes.